

Quantum Chemistry as a Typed Inference Stack

Many-electron states, reduced data, orbital gauge, nuclear geometry, and spectra

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Status. This document supersedes the bilingual working note v1.0 and is an application-layer reference migrated to GO Core v0.2. It does not introduce a new electronic-structure method, density functional, orbital interpretation, bond definition, or spectroscopic inversion theorem. Its internal contribution is a typed separation of the exact electron-nuclear state, clamped-nuclei electronic models, reduced density data, representation gauge, nuclear reduction, ideal observables, and finite-resolution measurement.

Abstract

Quantum chemistry is not described by a single composable chain $\Psi \rightarrow n \rightarrow E(R) \rightarrow R_* \rightarrow$ chemical behavior. Those arrows have different domains, hypotheses, and information content. This reference replaces that chain by a typed graph. The exact nonrelativistic molecular state depends on electronic and nuclear coordinates. A clamped-nuclei Hamiltonian family $H_e(R)$ generates potential-energy surfaces only after a Born-Oppenheimer reduction. For an N -electron density operator Γ_N , the one-body reduced density matrix $\gamma = N \text{Tr}_{2\dots N} \Gamma_N$ satisfies $\text{Tr} \gamma = N$, and the spatial density integrates to N , but this reduction is non-injective. Occupied-orbital rotations leave a Slater determinant ray and its one-body projector invariant, so localized orbitals, hybridizations, and active-space populations require an explicit representation protocol. At a stationary nuclear geometry, the mass-weighted Hessian has eigenvalues $\lambda_k = \omega_k^2$, not frequencies; five rigid zero modes for a linear molecule or six for a nonlinear molecule must be removed in the isolated, unconstrained case. An ideal transition spectrum depends on energies, populations, a transition operator, and a line profile, while measured data additionally depend on the instrument channel and noise. The resulting formalism preserves the useful observation-geometry interpretation but forbids claims of canonical orbital observability, unique bond projection, exact degeneracy from finite resolution, or chemistry determined by energy differences alone.

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1 Scope, state spaces, and unit firewall

Let N_e electrons and N_n nuclei be described in a declared inertial frame. Electronic spin coordinates are denoted by $x_i = (\mathbf{r}_i, \sigma_i)$, and nuclear Cartesian coordinates by $R = (\mathbf{R}_1, \dots, \mathbf{R}_{N_n})$. Ignoring relativity, external fields, nuclear spin, and finite nuclear size, the nonrelativistic Coulomb Hamiltonian in atomic units is

$$\begin{aligned} \hat{H}_{\text{mol}} = & -\frac{1}{2} \sum_{i=1}^{N_e} \nabla_i^2 - \sum_{A=1}^{N_n} \frac{1}{2M_A} \nabla_A^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} \\ & + \sum_{i<j} \frac{1}{r_{ij}} + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}. \end{aligned} \quad (1)$$

The nuclear masses M_A are expressed in electron-mass units. A full stationary state is

$$\Psi = \Psi(x_1, \dots, x_{N_e}; R_1, \dots, R_{N_n}), \quad \hat{H}_{\text{mol}} \Psi = E \Psi. \quad (2)$$

Writing only electronic arguments after equation (1) silently changes the problem and is therefore prohibited.

Boundary 1.1 (Unit context). Atomic units suppress $m_e, e, \hbar, 4\pi\epsilon_0$, but do not erase physical dimensions. A numerical atomic-unit expression may enter an SI formula only through an explicit conversion using the declared values of $E_h, a_0, m_e, e, \hbar$. No addition or comparison across SI and atomic contexts is implicit in this reference.

For pure isolated electronic states, the state space is the antisymmetric sector

$$\mathcal{H}_e^{(N)} = \bigwedge^N L^2(\mathbb{R}^3; \mathbb{C}^2).$$

Mixed or open-system electronic states are positive trace-one operators Γ_N on this sector. A wavefunction is therefore not the universal state object; it is a pure-state representative.

2 Exact molecule and Born-Oppenheimer reduction

At fixed nuclear geometry R , define the clamped-nuclei electronic Hamiltonian

$$\hat{H}_e(R) = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{en}(R) + V_{nn}(R). \quad (3)$$

This document includes the scalar nuclear repulsion $V_{nn}(R)$ in $\hat{H}_e(R)$; a convention that excludes it must add it back before calling an eigenvalue a potential-energy surface.

The parametric eigenproblem is

$$\hat{H}_e(R) \psi_k(x; R) = E_k(R) \psi_k(x; R). \quad (4)$$

The map $R \mapsto E_k(R)$ is not obtained by taking a marginal of the exact molecular wavefunction. It is generated from the Hamiltonian family after choosing a clamped-nuclei model and an electronic state label. A Born-Huang expansion has the form

$$\Psi(x, R) = \sum_k \chi_k(R) \psi_k(x; R). \quad (5)$$

Substitution into the full equation yields coupled nuclear equations containing derivative couplings. A one-surface Born-Oppenheimer model is a reduction obtained by neglecting specified off-diagonal terms; it is not a lossless frame transform and it is not uniformly valid at a closing internal electronic gap.

Boundary 2.1 (Conical-intersection inheritance). Near a conical intersection, the individual rank-one adiabatic projectors need not extend through the seam even when the retained two-state cluster remains regular. Branching coordinates, gauge, Berry holonomy, derivative coupling, dynamics, and finite-resolution identifiability are governed by the separate P8 conical-intersection contract. The present document does not replace that contract with the phrase “observation caustic.”

3 Reduced density data and information loss

Definition 3.1 (One-body reduced density matrix). For a normalized N -electron state Γ_N , define

$$\gamma^{(1)} = N \operatorname{Tr}_{2,\dots,N} \Gamma_N. \quad (6)$$

Its spatial density is

$$n(\mathbf{r}) = \sum_{\sigma} \langle \mathbf{r}\sigma | \gamma^{(1)} | \mathbf{r}\sigma \rangle. \quad (7)$$

Proposition 3.2 (Normalization and one-body observables). *The reduction in equation (6) obeys*

$$\operatorname{Tr} \gamma^{(1)} = N, \quad \int_{\mathbb{R}^3} n(\mathbf{r}) \, d^3r = N. \quad (8)$$

For every one-body observable $\hat{A} = \sum_{i=1}^N a(i)$,

$$\operatorname{Tr}(\Gamma_N \hat{A}) = \operatorname{Tr}(\gamma^{(1)} a). \quad (9)$$

Proof. The partial trace preserves the trace, so $\operatorname{Tr} \gamma^{(1)} = N \operatorname{Tr} \Gamma_N = N$. Resolving the one-particle trace in the position-spin basis gives the density integral. Permutation symmetry of Γ_N makes all N terms in $\hat{A}$ equal after tracing the other particles, which yields equation (9). $\square$

Proposition 3.3 (The one-body reduction is non-injective). *Let $|ij\rangle = a_i^\dagger a_j^\dagger |0\rangle$ in $\wedge^2 \mathbb{C}^4$, and define*

$$|\Psi_{\pm}\rangle = \frac{|12\rangle \pm |34\rangle}{\sqrt{2}}.$$

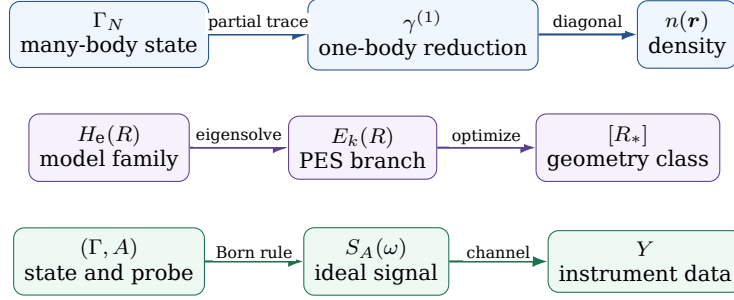
Both states have $\gamma^{(1)} = \frac{1}{2} I_4$, but the two-body coherence

$$\hat{B} = |12\rangle\langle 34| + |34\rangle\langle 12|$$

has expectation +1 in Ψ_+ and -1 in Ψ_- .

Proof. Each spin orbital occurs with probability 1/2, and all one-body cross terms vanish because the determinants differ by two orbitals. The operator $\hat{B}$ exchanges the two determinants, producing the stated signs. $\square$

Thus neither $\Gamma_N \mapsto \gamma^{(1)}$ nor $\Gamma_N \mapsto n$ is generally invertible. Electron correlation is not defined as an information-theoretic defect of this map. It is method- and Hamiltonian-specific unless a precise correlation functional or entanglement quantity is declared.



The three rows are related by a declared scientific workflow, but they are not a single automatic composition.

4 Finite bases, Hartree-Fock, and orbital gauge

Let $\{\chi_\mu\}_{\mu=1}^K$ be a linearly independent, generally nonorthogonal one-particle basis with overlap matrix

$$S_{\mu\nu} = \langle \chi_\mu, \chi_\nu \rangle, \quad S = S^\dagger > 0.$$

Roothaan–Hall equations have the generalized form

$$FC = SC\varepsilon, \quad C^\dagger SC = I. \quad (10)$$

Writing $FC = C\varepsilon$ without either an orthonormal basis or the overlap matrix changes the discrete problem. Small eigenvalues of S also require a declared linear-dependence threshold.

For occupied columns C_o , define the AO density matrix

$$P = C_o C_o^\dagger, \quad \text{Tr}(PS) = N_{\text{occ}} \quad (11)$$

for one electron per occupied spin orbital. Closed-shell spatial orbital conventions add their declared occupation factor.

Proposition 4.1 (Occupied-orbital gauge). *If $U \in U(N_{\text{occ}})$, then*

$$C'_o = C_o U \implies P' = P.$$

The associated normalized Slater determinant changes only by $\det U$, hence represents the same quantum ray.

Proof. $P' = C_o U U^\dagger C_o^\dagger = P$. Multilinearity of the exterior product gives $\phi'_1 \wedge \cdots \wedge \phi'_{N_{\text{occ}}} = (\det U) \phi_1 \wedge \cdots \wedge \phi_{N_{\text{occ}}}$, and $|\det U| = 1$. $\square$

Consequently, localized occupied orbitals, hybrid orbitals, and orbital-by-orbital bond pictures are representation choices unless the localization or decomposition functional is part of the protocol. Canonical orbitals diagonalize a chosen effective operator; that gauge choice does not turn every orbital or orbital energy into a many-body observable.

Hartree–Fock is the variational problem

$$E_{\text{HF}} = \inf_{\Phi \text{ Slater}, \|\Phi\|=1} \langle \Phi | \hat{H}_e | \Phi \rangle. \quad (12)$$

The nonlinear SCF equations characterize stationary solutions. An iteration may converge to a local stationary point, oscillate, or break an exact symmetry; convergence alone does not certify the global Hartree-Fock minimum.

For the same Hamiltonian and one-particle space,

$$E_{\text{corr}} = E_0 - E_{\text{HF}} \leq 0, \quad (13)$$

where E_0 is the exact or full-CI ground energy in the declared space. Cross-method correlation energies are not comparable unless Hamiltonian, basis, frozen-core convention, geometry, and reference state agree.

5 Density-functional theory firewall

The density map is a partial trace followed by a diagonal readout. Ground-state density-functional theory adds a separate variational statement. For fixed particle number and electron-electron interaction, a constrained-search functional can be written

$$F[n] = \inf_{\Gamma \mapsto n} \text{Tr} \left[\Gamma (\hat{T} + \hat{W}) \right]. \quad (14)$$

The external-potential ground energy is then

$$E_v = \inf_n \left\{ F[n] + \int v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \, \mathrm{d}^3 r \right\}, \quad (15)$$

over the declared admissible density class.

Boundary 5.1 (Hohenberg-Kohn scope). In its original ground-state setting, the density determines the external scalar potential up to an additive constant under fixed particle number and interaction, subject to the theorem’s representability and degeneracy hypotheses. This does not invert the density map for an arbitrary excited, time-dependent, mixed, or open-system state. Degenerate and ensemble cases require their own formulation.

The Kohn-Sham construction introduces auxiliary orbitals with

$$n(\mathbf{r}) = \sum_i f_i |\varphi_i(\mathbf{r})|^2, \quad \left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}[n] \right] \varphi_i = \varepsilon_i \varphi_i \quad (16)$$

in atomic units. The orbitals reproduce the target density within the model. General Kohn-Sham eigenvalue differences are not equal by definition to interacting excitation energies, and approximate exchange-correlation functionals introduce model error not encoded by the density normalization.

6 Nuclear geometry and the normal-mode theorem

For an isolated molecule, absolute translation and rotation do not define internal geometry. Away from collisions and singular configurations, an internal configuration is an equivalence class

$$[R] \in \mathcal{Q}_{\text{int}} = ((\mathbb{R}^{3N_n} \setminus \Delta) / \text{SE}(3)) / \Pi_{\text{identical}},$$

where the final quotient is included when identical-nucleus permutations are physically identified. A Cartesian minimizer is therefore not unique before this quotient or a gauge-fixing alignment.

Let R_* be a stationary geometry of a smooth PES, $H_R = \nabla_R^2 E(R_*)$ its Cartesian Hessian, and $\mathbf{M} = \text{diag}(M_1, M_1, M_1, \dots)$. The mass-weighted dynamical matrix is

$$\mathbf{K} = \mathbf{M}^{-1/2} H_R \mathbf{M}^{-1/2}. \quad (17)$$

Theorem 6.1 (Harmonic normal modes). *If $\mathbf{K}u_k = \lambda_k u_k$, then the linearized nuclear equation has modal solutions with*

$$\lambda_k = \omega_k^2, \quad \nu_k = \frac{\omega_k}{2\pi}, \quad \tilde{\nu}_k = \frac{\omega_k}{2\pi c}. \quad (18)$$

For an unconstrained isolated nonlinear molecule, six rigid translation/rotation directions are removed; for a linear molecule, five are removed. A strict harmonic minimum has $\lambda_k > 0$ on the remaining vibrational subspace. A first-order transition state has exactly one negative vibrational eigenvalue there.

Proof. With displacement δR , the linearized equation is $\mathbf{M}\delta\ddot{R} + H_R\delta R = 0$. Setting $q = \mathbf{M}^{1/2}\delta R$ gives $\ddot{q} + \mathbf{K}q = 0$. Substitution $q(t) = u_k e^{i\omega t}$ yields $\mathbf{K}u_k = \omega^2 u_k$. Euclidean invariance makes translations and infinitesimal rotations null directions at a stationary geometry. Rotation about the molecular axis displaces no nucleus for a linear geometry, leaving five rather than six independent rigid directions. $\square$

Boundary 6.2 (Qualification). Atoms, constraints, periodic cells, external fields, nonstationary points, exact symmetries beyond rigid motion, and numerical finite-difference Hessians change the null-space analysis. A small computed value is not automatically an exact zero mode; the threshold must be scaled and reported.

7 Active spaces, valence, and bond interpretations

A mathematically typed active-space reduction begins with a declared one-particle projector P_A :

$$\gamma_A = P_A \gamma^{(1)} P_A, \quad N_A = \text{Tr}(\gamma^{(1)} P_A). \quad (19)$$

The projector may be generated by atomic valence functions, energy windows, natural occupations, localized orbitals, or another stated algorithm. Different admissible choices need not give the same N_A , oxidation assignment, or bond decomposition.

Definition 7.1 (Interpretation map). A population, partial charge, bond order, hybridization, or “valence projection” is an interpretation map

$$\mathfrak{I}_\alpha : (\Gamma, H, \mathcal{B}, \mathcal{A}, \text{conventions}) \mapsto z_\alpha$$

with a named scheme α , basis $\mathcal{B}$, active space $\mathcal{A}$, and normalization. It is not promoted to a scheme-independent observable without an invariance theorem.

This distinction does not make such quantities useless. It makes comparisons conditional: trends are meaningful within a controlled protocol, while equality across unrelated partition schemes is not assumed.

8 Ideal spectra and finite-resolution data

Let $\{|i\rangle, E_i\}$ be stationary states, p_i initial populations, and $\hat{A}$ the probe-dependent transition operator. With a normalized line profile $\int L_\Gamma(\omega) d\omega = 1$, define

$$S_A(\omega) = \sum_{i,f} p_i \left| \langle f | \hat{A} | i \rangle \right|^2 L_\Gamma \left(\omega - \frac{E_f - E_i}{\hbar} \right). \quad (20)$$

The observed record is a separate channel, schematically

$$Y = \mathcal{D}_\epsilon \mathcal{C}_{\text{inst}}[S_A] + \eta, \quad (21)$$

where $\mathcal{C}_{\text{inst}}$ includes response and convolution, $\mathcal{D}_\epsilon$ sampling or quantization, and η a declared stochastic error process.

Energy differences determine line locations only after a dynamical model identifies the relevant states. Intensities also require populations and matrix elements. Broadening, unresolved blends, calibration, baseline, and selection bias enter the inference from Y back to a molecular model.

If a symmetry product contains the totally symmetric representation, a transition is symmetry-allowed for the declared operator. This is a necessary symmetry condition, not a guarantee that the matrix element is numerically nonzero: additional quantum numbers, cancellation, population, or experimental geometry may suppress the line.

9 Approximation and identifiability ledger

A reproducible quantum-chemical claim separates at least the following errors:

1. **physical-model error:** nonrelativistic Hamiltonian, nuclear model, environmental and field terms;
2. **electron-nuclear reduction error:** Born-Oppenheimer, retained surfaces, derivative couplings, nuclear quantum effects;
3. **discretization error:** one-particle basis, active space, grid, pseudopotential, boundary conditions;
4. **many-body model error:** determinant ansatz, excitation rank, density functional, embedding approximation;
5. **numerical error:** SCF stationarity, integral and diagonalization tolerances, geometry and Hessian convergence;
6. **property-model error:** response order, transition operator, gauge, relativistic picture-change corrections;
7. **observation error:** instrument response, resolution, calibration, noise, estimator, and assignment ambiguity.

Agreement of one final scalar with experiment does not identify these components separately.

Boundary 9.1 (Relativistic boundary). For heavy elements, a relativistic model must state its nuclear charge distribution, chosen positive-energy or no-pair space, two-electron terms, and finite-basis kinetic balance. An unqualified nonrelativistic Rayleigh-Ritz argument cannot be transferred to the Dirac operator across its negative-energy continuum.

10 Reproducible controls and claim register

The strict program-level result is a typed inference graph: Hamiltonian selection, state preparation, many-body reduction, representation choice, nuclear reduction, ideal property evaluation, and measurement are distinct operations. Structural analogy is retained only where their domains and invariants are explicit.

Control	Exact result	Purpose
$ \Psi_{\pm}\rangle = (12\rangle \pm 34\rangle)/\sqrt{2}$	$\gamma^{(1)} = \frac{1}{2}I_4, \langle B \rangle = \pm 1$	Non-injectivity of one-body reduction
$C_0 \mapsto C_0 U$	$P' = P$, Slater ray unchanged	Occupied-orbital gauge
Diatomic spring k , masses m_1, m_2	$\omega^2 = k(m_1^{-1} + m_2^{-1})$	Frequency-square and rigid-mode test
Two-determinant matrix $\begin{pmatrix} 0 & V \\ V & \Delta \end{pmatrix}$	$E_0 = (\Delta - \sqrt{\Delta^2 + 4V^2})/2 \leq 0$	Variational correlation sign
Gaussian L_{σ}	$\int L_{\sigma}(\omega - \omega_0) d\omega = 1$	Line-profile normalization
Nested Ritz spaces	$E_0^{(K+1)} \leq E_0^{(K)}$	Finite-basis variational control

Table 1: Reference values implemented by the P9 regression suite.

Claim	Status	Scope
$\text{Tr} \gamma^{(1)} = N$ and $\int n = N$	Proposition	Normalized N -electron state
One-body reduction is non-injective	Proposition	Explicit two-electron four-orbital control
Occupied rotations preserve P	Proposition	Unitary rotations within occupied space
$\lambda_k = \omega_k^2$	Theorem	Harmonic mass-weighted Hessian
Five or six rigid zero modes	Qualified theorem	Isolated linear or nonlinear molecule
Density uniquely reconstructs any state	Rejected	Hohenberg-Kohn is a ground-state theorem
Orbitals or bonds are unique observables	Rejected	Representation and scheme must be declared
Energy differences alone determine a spectrum	Rejected	Probe, populations, line shape, and channel are required
New quantum-chemical method	Rejected	Not claimed

Protocol minimum

A complete record states at least: particle numbers and isotope masses; Hamiltonian and unit context; frame, geometry convention, and symmetry; pure or mixed state; electronic and nuclear reduction; basis, overlap threshold, active space, and frozen sectors; orbital gauge or localization rule; correlation or density-functional model; SCF and geometry stationarity criteria; Hessian projection and zero-mode threshold; property operator and gauge; populations, line profile, instrument response, noise law, estimator, uncertainty, and identifiability scope.

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